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JUNE 28, 1960
SUPERSEDING
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May 2, 1957

FEDERAL SPECIFICATION

MARKER SET, X-RAY FILM IDENTIFICATION

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope.—This specification covers X-ray film marker sets, suitable for use in identifying X-ray films. This specification also includes accessories, such as holders and storage boxes.

1.2 Classification.

1.2.1 *Classes, sizes, and sets.*—Letters and numerals shall be of the following classes, sizes, and sets, as specified (see 6.1) :

Class 1.—Hairline gothic, tapers to a sharp edge at top.

Class 2.—Sharp-face gothic, tapers toward top, has flat surface.

Class 3.—Flat, gothic style, capital letters.

Size 1.—¼ inch ± 1/64 inch high.

Size 2.—½ inch ± 1/64 inch high.

Size 3.—¾ inch ± 1/64 inch high.

Size 4.—1 inch ± 1/64 inch high.

Set A.—One complete alphabet A to Z, inclusive.

Set B.—One each of letters R and L.

Set C.—One complete set of numerals 0 to 9, inclusive.

Set D.—One each of letters, F, S, P, R, and L.

Set E.—Fifteen complete sets of alphabets A to Z, inclusive.
Twenty each of letters R and L.
Forty complete sets of numerals 0 to 9.
Twenty-five each of letters A, B, C, D, E, F, G, J, L, M, N, O, P, R, S, T, U, V, and Y.

Set F.—One set containing 12 each of the same letters or numerals.

1.2.2 *Holders.*—Holders shall be of the following types, as specified (see 6.1) :

Type A.—Single channel holder.

Type B.—Double channel holder.

1.2.3 *Storage boxes.*—Storage boxes shall be of the following styles, as specified (see 6.1) :

Style a.—Sliding cover.

Style b.—Hinged cover.

2. APPLICABLE SPECIFICATIONS AND STANDARDS

2.1 Specifications and standards.—The following specifications and standards, of the issues in effect on date of invitation for bids, form a part of this specification:

Federal Specifications:

QQ–A–561—Aluminum Alloy 1100 (2S) ; Plate and Sheet.

QQ–L–171—Lead; Pig.

QQ–L–201—Lead; Sheet.

PPP–B–566 — Boxes, Folding, Paperboard.

PPP–B–585—Boxes, Wood, Wirebound.

PPP–B–601—Boxes, Wood, Cleated-Plywood.

PPP–B–621—Boxes, Wood, Nailed and Lock-Corner.

PPP–B–636—Box, Fiberboard.

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PPP–B–665—Boxes; Paperboard, Metal Stayed (Including Stay Material).

PPP–B–676—Boxes, Set-Up, Paperboard.

Federal Standards:

Fed. Std. No. 102—Preservation, Packaging, and Packing Levels.

Fed. Std. No. 123—Marking for Domestic Shipments (Civilian Agencies).

Fed. Std. No. 595—Colors.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications, Standards, and Handbooks and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, Seattle, and Washington, D. C.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications, Standards, and Handbooks from established distribution points in their agencies.)

Military Specification:

MIL–L–10547 — Liners, Case, Waterproof.

Military Standards:

MIL–STD–105 — Sampling Procedures and Tables for Inspection by Attributes.

MIL–STD–129—Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by the contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material.—Satisfactory high-grade materials shall be used throughout. All parts entering into the construction of the X-ray marker set covered by this specification shall be manufactured from materials which are entirely satisfactory for the purpose and of sufficient strength to give satisfactory performance for years of severe service.

3.1.1 *Aluminum.* — Aluminum shall conform to Federal Specification QQ–A–561, for H14 temper, having a corrosion-resisting coating produced by nonelectrolytic chemical process (see 6.3).

3.1.2 *Lead.*—Lead shall conform to Federal Specification QQ–L–201 for grade C lead, if letters and numerals are stamped, and to Federal Specification QQ–L–171 for grade A lead, if letters and numerals are molded. In order for the figures to cast properly, the total tin, arsenic, antimony, and copper content of the lead may be increased but shall not exceed 6 percent.

3.1.3 *Plastic.* — Plastic material shall be bakelite or cellulose acetate of the nonwarping and slow burning type.

3.1.4 *Wood.*—Wood shall be of aspen, basswood, buckeye, butternut, chestnut, cottonwood, yellow poplar, Port Orford white cedar, western red cedar, redwood, ponderosa pine, sugar pine, eastern white pine, or western white pine. Wood shall be free from defects other than surface checks, tight small knots, light pitch, fine shake, and medium stain.

3.2 Letters and numerals.

3.2.1 *Count.*—The quantity of letters and numerals required under a contract shall consist of a minimum of the total quantity of characters specified. Individual groups of letters or numerals shall have no greater than one less than the number of characters specified. Overages are acceptable.

3.2.2 *Design and construction.* — Letters and numerals shall be 0.0391 ± 0.005 inch thick, of the sizes specified (see 1.2), and of a width suitably proportionate to their height. The letters and numerals shall be stamped or molded of lead as specified in 3.1.2.

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3.2.2.1 *Mounting.*—Unless otherwise specified, the letters and numerals shall be mounted (see 6.1). The letters and numerals shall be mounted with a cement or glue which will adhere firmly to both the lead and plastic strip. The mounting shall be carefully done so that the cement will not interfere when sliding the letters and numerals in the holder. The plastic backing shall be as specified in 3.1.3. The plastic-backed letters and numerals shall be considered nonwarping if they fit into an appropriate holder without changing their shape or buckling.

3.2.2.1.1 *Adherence.*—The adhesive used to attach the lead letters and numerals to the plastic backing shall be of superior quality. The lead letters and numerals shall not separate, nor shall the lead letters and numerals snap off when tested in accordance with 4.4.3.

3.2.2.1.2 *Straightness.*—All characters shall be vertical within 1/64 inch from absolute vertical. From an established line, approximately 3/16 inches from the bottom of the plastic, all characters shall be within 1/64 inch from absolute horizontal. All measurements shall be made with the characters in an appropriate holder.

3.2.2.2 *Size 1 and size 2 letters and numerals.*—Size 1 and size 2 letters and numerals shall be mounted in the center of a durable plastic strip 0.625 inch ± 0.010 inch high, 5/16 inch ± 1/32 inch wide, and 0.035 ± 0.005 inch thick.

3.2.2.2.1 *Size 3 and size 4 letters and numerals.*—Size 3 and size 4 letters and numerals shall be mounted in the center of a durable plastic strip 1 1/4 inches ± 0.010 inch high, 5/8 inch ± 1/32 inch wide, and 0.035 ± 0.005 inch thick.

3.2.3 *Holders, type A and type B.* — The channels of the type A and type B holders shall be designed to hold the 0.040 inch thick plastic-backed numerals and letters and to permit them to slide into place easily. For size 1 and size 2 letters and numerals, the holders shall be 4 1/2 ± 1/8 inches long. For size 3 and size 4 letters and numerals, the holders shall be 9 inches ± 1/4 inch long. The holders shall be fabricated of sheet aluminum, as specified in 3.1.1, and the aluminum shall be a minimum of 0.018 inch thick. The maximum thickness of the marker, with letters and numerals in place, shall not exceed 0.125 inch.

3.2.3.1 *Type B, double-channel holder.* — For size 1 and size 2 letters and numerals, the double-channel holder shall be not more than 1 7/16 inches wide. For size 3 and size 4 letters and numerals, the double-channel holder shall be not more than 2 7/8 inches wide. The double-channel holders shall be so formed as to be 2 channels, each arranged to accommodate the letters and numerals specified in 3.2.2. The channels of each double channel holder shall be held in place by spot welding.

3.2.4 *Storage boxes.*

3.2.4.1 *Style a, sliding cover.* — Style a, sliding cover storage box shall be made of wood as specified in 3.1.4. It shall be made of approximately 1/4-inch material. The box shall be divided into 36 compartments of equal size for storing the letters and numerals. Space shall also be provided for storing one double-channel holder. The joints of the box shall be dovetailed or of similar construction. The corners of the box shall be glued, and the ends of the strips forming the compartments shall be securely glued to the sides of the box. The bottom of the box shall be secured by nails. A groove approximately 3/16 inch deep shall be formed on the upper inner surface of both sides and back. The front end of the groove shall be open to receive a 1/4 inch thick sliding cover of the same material. The cover shall contain a fingernail groove to facilitate opening and closing.

3.2.4.2 *Style b, hinged cover.* — Style b, hinged cover box may be fabricated of wood as specified in 3.1.4, metal, or plastic. There shall be 36 compartments for letters and numerals and a space for two double-channel

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holders. When closed, the overall dimensions of the storage box shall be 9½ inches by 12½ inches by 2 inches (± ½ inch tolerance on each dimension). The compartments shall be 1⅜ inch by 1⅜ ± ⅛ inch tolerances. The holder area shall have a width of 1⅞ ± ⅛ inch. The paneling between the compartments shall be ⅛ ± 1/32 inch thick. Each compartment shall have one side wall curved to facilitate removing the lead letters and numerals. Construction of the compartments shall be sturdy and shall be securely fastened to prevent accidental removal of partitions. A letter or numeral shall either be permanently mounted or stamped on the back of each compartment to indicate the proper compartment for each letter or numeral. These letters and numerals shall be easily visible when the lid is open and the proper number of letters or numerals are in each compartment. The box shall have a hinged lid lined with sponge rubber, cemented in place, and so built that when the lid is closed, it will not be possible for lead numerals or letters to move to adjacent compartments when the box is turned upside down or shaken in any position. The lid shall open approximately 105° and shall be so designed that it will remain open when placed in that position. The hinges shall be sturdy and firmly mounted on either the short or the long side of the box. A spring-action catch shall be provided to close the lid.

3.3 Finish.—The style b, hinged cover box, as specified in 3.2.4.2, shall be finished with a smooth durable enamel, utilizing a suitable primer coating. The color of the enamel shall be semi-gloss green, color 24087, as shown in Fed. Std. No. 595 (see 6.2). Plastic boxes shall be moulded in the referenced color and polished. If required, the finish of style a, sliding cover box, as specified in 3.2.4.1, shall be designated (see 6.1).

3.4 Marking.—When a storage box complete with letters and numerals is supplied, the cover of the storage box shall be permanently and legibly marked with the name or registered trademark of the manufacturer of the lead letters and numerals, the contract number, stock number, and the letters "U.S."

3.5 Workmanship. — Workmanship shall be first class throughout. The marker sets shall be free from defects which detract from their appearance or which may impair their serviceability.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Supplier responsibilities for inspection. —Unless otherwise specified herein, the supplier is responsible for the performance of all inspection requirements prior to submission for Government inspection and acceptance. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. Inspection records of the examinations and tests shall be kept complete and available to the Government as specified in the contract or order.

4.1.1 Inspection, as used in this specification, is defined as both examination (such as visual, auditory, etc., investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.2 *Certificates of quality.*—When available, certificates of quality, supplied by the manufacturer of the component or material, listing the specified test method and test results obtained, may be furnished in lieu of actual performance of such testing by the contractor.

4.2 Sampling. — Sampling shall be conducted in accordance with Military Standard MIL–STD–105. The unit of product for inspection purposes shall be divided into two parts, namely, storage boxes and letter and numeral sets. Inspection shall be conducted separately on storage boxes and letter and numeral sets, as applicable. For letter and numeral sets, the unit of product for inspec-

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tion purposes shall be each like letter or numeral included in the set.

4.2.1 *Marker sets.*

4.2.1.1 *Storage boxes.*—The lot size of the storage boxes shall be determined by the quantity existing in the prime lot of sets submitted at inspection level L2.

4.2.1.2 *Letters and numerals.*—Letters and numerals included in the marker sets shall be treated separately in accordance with the instructions for the letter and numeral sets, as indicated below.

(a) Letter and numeral sets in the prime lot submitted shall be subdivided into sublots. Each sublot shall consist of like letters or numerals only.

(b) The sublot size for each letter or numeral shall be determined from the number of like letters or numerals in the prime lot of sets submitted.

(c) The number of sample letters and numerals to be selected at random from each sublot shall be determined from table V in Military Standard MIL–STD–105 for reduced inspection on a single sampling basis only.

4.2.2 *Criteria for acceptance or rejection.*—For marker sets, acceptance of the prime lot of material submitted shall be subject to acceptance of the component parts included in the set. For prime letter and numeral sets, the disposition of each sublot shall be determined in accordance with the appropriate acceptance or rejection numbers indicated in table V of Military Standard MIL–STD–105 for acceptable quality levels (AQL's) specified herein. In the event that one or more sublots of letters or numerals is rejected, the overall prime lot shall be rejected, except that upon satisfactory repair or replacement of the defective sublots, as verified by reinspection, the prime lot shall be considered acceptable.

4.3 **Examination.**—The marker set shall be examined to determine compliance with all requirements contained in this specification.

4.3.1 *Classification of defects.*—Examination of letters and numerals and marker sets shall be conducted in accordance with the following classification of defects:

*Categories and defects**

Unit of Product—Letters and numerals

Major A

101. Letters and numerals not of class, size, or set specified.
102. Dimensions of plastic, and letters and numerals not as specified.
103. Mounted letters and numerals do not slide easily into appropriate holder.

Minor A

201. Mounted letters and numerals not centered and vertical in plastic backing as described in 3.2.2.1.2.

Unit of Product—Holders

Major B

125. Holders not of types and dimensions specified.
126. Holders not free of sharp corners, sharp edges, cracks, or deformities.
127. Spotwelds on type B channel holder not sound and not free from pits, deformed, or corrosive areas.

Unit of Product—Storage boxes

Major C

150. Storage boxes not of style, design, or dimensions specified.
151. Storage boxes not free from burs, slivers, sharp edges, or deformities.
152. Storage boxes not compartmented as specified.
153. Each compartment not permanently or legibly marked to readily indicate letter or numeral for selection.
154. Letter and numeral compartments not constructed to facilitate the removal of these letters and numerals as specified (see 3.2.4.2).
155. Letters and numerals move to adjacent compartments when storage box is turned upside down and shaken.
156. Lid of style b storage box does not remain open at approximately 105°.

Minor C

251. Marking of storage boxes not as specified, permanent, or legible.
252. Finish of style b storage box not as specified.

*Examination shall not be restricted to the classified possible defects listed above.

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4.4 Tests.—Tests shall be conducted, as necessary, to determine compliance with specification requirements.

4.4.1 *Chemical analysis.*—Four ounces of each metal used in the construction of the letters and numerals, and holders, shall be selected for chemical analysis (see 4.1.2).

4.4.2 *Count.*—Five marker sets per thousand or fraction thereof containing letters and numerals shall be selected at random from each prime lot. The count shall meet the requirements specified in 3.2.1.

4.4.3 *Adherence test.*—An average of 10 letters or numerals shall be selected at random from each marker set selected in the lot. Five of the letters or numerals shall be dropped to a concrete floor at a distance of 4 feet and shall meet the requirements of 3.2.2.1.1. Each end of the plastic backing of the other five letters or numerals shall simultaneously be bent backward through angles of 20° from a tangential to the center of the backing and shall meet the requirements of 3.2.2.1.1. If destruction during bend occurs prior to the bend angles specified and the letters or numerals have not snapped off, the adhesive shall be considered satisfactory.

5. PREPARATION FOR DELIVERY

(For civil agency procurement Federal Standard No. 102 shall be referred to for definitions and applications of the various levels of packaging and packing protection for supplies and equipment.)

5.1 Packaging.

5.1.1 *Unit of issue.*—One marker set, as specified, constitutes one unit.

5.1.2 *Quantities.*—Packaging and packing quantities shall be as specified in table I.

5.1.2.1 *Packing variation permitted.* — If the required number of units to be shipped is less than the number of units specified to be overpacked in an exterior container, such units shall be packed in an exterior container of suitable size and design, acceptable to a common carrier, which will insure safe delivery to destination.

TABLE I.—*Packaging and packing quantities*

Set	Unit package	Intermediate package	Exterior container
	Units	*Units*	*Units*
A	1	12	576
B	1	144	3,456
C	1	12	576
E	1	Not required	12

5.1.3 *Level A.*

5.1.3.1 *Unit package.*

5.1.3.1.1 *Set A.*—Each unit shall be packaged in a carton of appropriate size constructed in accordance with Federal Specification PPP–B–566, PPP–B–665, or PPP–B–676. Carton shall be designed with suitable interior fittings or shall contain sufficient cushioning material to insure against shifting of, or damage to, the contents. Carton shall be adequately secured to prevent accidental opening.

5.1.3.1.2 *Set B or set C.*—Each unit shall be packaged in an envelope of suitable size and design. Envelope shall be adequately secured to prevent accidental opening.

5.1.3.1.3 *Set E.*—Each unit shall be packaged in a carton of appropriate size constructed in accordance with Federal Specification PPP–B–636 or PPP–B–676. Carton shall be adequately secured to prevent accidental opening.

5.1.3.2 *Intermediate package.*

5.1.3.2.1 *Set A.*—The number of units specified in table I shall either be packaged in a carton of appropriate size and design, or

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shall be completely wrapped with kraft paper of suitable weight. Carton or wrapping shall be adequately secured to prevent accidental opening.

5.1.3.2.2 *Set B.*—The number of units specified in table I shall be packaged in a carton of appropriate size constructed in accordance with Federal Specification PPP–B–636, type I, class 1. Carton shall be adequately secured to prevent accidental opening.

5.1.3.2.3 *Set C.*—The number of units specified in table I shall be packaged in a carton of appropriate size constructed in accordance with Federal Specification PPP–B–566, PPP–B–665, or PPP–B–676. Carton shall be adequately secured to prevent accidental opening.

5.1.3.2.4 *Set E.* — Intermediate packaging is not required.

5.1.4 *Level C sets A, B, C, D, E, or F.*— Unless otherwise specified, commercial packages shall be acceptable.

5.2 Packing.

5.2.1 *Level A.*

5.2.1.1 *Exterior container sets A, B, C, or E.*—The number of units specified in table I shall be overpacked in an exterior container designed for type 2 load and constructed in accordance with Federal Specification PPP–B–585, style 3, for class 3 use; PPP–B–601, table II, using type III, class 1 plywood; PPP–B–621, table III.

5.2.1.2 *Case liner.*—Each level A box shall be lined with a waterproof case liner conforming to Military Specification MIL–L–10547. Closure and sealing shall conform to applicable paragraphs of appendix thereto.

5.2.1.3 *Strapping.* — Strapping, when required, shall be in accordance with applicable box specification.

Note.—Strapping will not be required for shipment forwarded to a receiving activity within the continental limits of the United States for storage and redistribution.

5.2.2 *Level B.*

5.2.2.1 *Exterior container—Set A, B, C, or E.*—The number of units specified in table I shall be overpacked in an exterior container conforming to Federal Specification PPP–B–601, table I, using type III, class 1 plywood, or shall be designed for type 2 load and constructed in accordance with Federal Specification PPP–B–585, style 3, for class 2 use; PPP–B–621, table I; or PPP–B–636, type I, class 1, fabricated from not less than 275-pound test fiberboard.

5.2.3 *Level C—Set A, B, C, D, E, or F.*— The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.3 Marking.

5.3.1 *For civil agencies.*—Interior packages and shipping containers shall be marked in accordance with Federal Standard No. 123.

5.3.2 *For military agencies.*

5.3.2.1 *Unit package.*

5.3.2.1.1 *Set A, Set B, or Set C.*—Each unit package shall be marked with stock number, item identification, unit, quantity, and name of contractor.

5.3.2.1.2 *Set E.*—Each unit package shall be marked in accordance with section 5 of Military Standard MIL–STD–129.

5.3.2.2 *Intermediate package — Set A, Set B, or Set C.* — Each intermediate package shall be marked in accordance with section 5 of Military Standard MIL–STD–129.

5.3.2.3 *Exterior container—Set A, B, C, or E.* — Exterior container shall be marked in accordance with Military Standard MIL–STD–129.

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6. NOTES

6.1 Ordering data.—Purchasers should exercise any desired options offered herein, and procurement documents should specify the following:

(*a*) Title, number and date of this specification.
(*b*) Class, size, and set required (see 1.2).
(*c*) When unmounted letters or numerals are required (see 3.2.2.1).
(*d*) Type of holder and the quantity, when required (see 1.2.2).
(*e*) Style of storage box, when required (see 1.2.3).
(*f*) Finish of storage box, style a, if required (see 3.3).
(*g*) Selection of applicable level of packaging and packing (see 5.1 and 5.2).

6.2 Availability of color chips.—The individual 3- by 5-inch chip of the color referenced in 3.3 is available from General Services Administration, Business Service Center, Region 3, Seventh and D Streets, S.W., Washington 25, D. C. Price 5 cents.

6.3 Chemical finish.—The alodine process is considered a satisfactory process for the chemical finish of the aluminum (see 3.1.1).

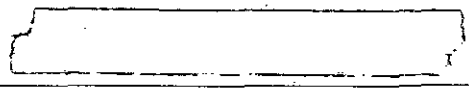
6.4 This specification does not include all types, classes, grades, sizes, etc., of the commodity indicated by the title of the specification, or which are commercially available, but is intended to cover the type, etc., normally procured to meet Federal Government requirements.

6.5 Transportation description.—Transportation description applicable to this item is:

Letters, Alphabetical or Numerical, NOIBN
Carload minimum weight, 30,000 pounds.
Motor volume minimum weight 20,000 pounds.

Notice. — When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded, by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

CUSTODIANS:
Army—Army Medical Service
Navy—Bureau of Medicine and Surgery
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